to install and maintain software, with updates managed by the provider. It's widely used for everything from email to customer relationship management (CRM). Examples include Salesforce, Google Workspace, and Microsoft 365.

How cloud technologies are viewed

End user's (customer's) perspective: Cloud technologies improve overall experience, accessibility, and ease for end users. If users have an internet connection, cloud-based applications let them access data and services from any device, anywhere in the globe. This implies that they won't lose data when switching between devices. The cloud also facilitates security updates and does so automatically, guaranteeing that customers always have access to the newest features without the need for manual intervention. Customer engagement and satisfaction are greatly increased by this dependability and convenience of use.

Developer's perspective: Cloud technologies are revolutionary from the standpoint of a developer. Without having to worry about the underlying infrastructure, developers can build, test, and deploy apps fast in the cloud's scalable and adaptable environment. PaaS tools streamline the development process so that developers can concentrate on developing code instead of administering servers. Additionally, the cloud makes it easier for distant teams to collaborate and work on multiple projects at once. Developers can innovate more quickly and incorporate cutting-edge features like artificial intelligence (AI), machine learning, and Big Data analytics into their apps with the help of strong cloud-based tools and APIs.

Organisation's perspective: Cloud technology provides businesses with several operational and financial benefits. Businesses can cut expenses related to maintaining physical IT infrastructure, such as servers and data centres, by implementing cloud solutions. They may optimise their IT spending by scaling resources up or down in response to demand with the pay-as-you-go concept. Faster time-to-market is also made possible by cloud computing, since businesses can expand their current services or launch new ones without having to wait for lengthy lead times. Furthermore, without having to have a physical presence in every location, businesses may extend their operations and deliver services to clients around the globe thanks to the global accessibility of cloud providers.

From the perspective of environmental sustainability: Cloud computing helps maintain the environment. Cloud providers can lower overall energy usage compared to traditional on-premises data centres by grouping computing resources into large, energy-efficient data centres. To reduce their carbon footprint, these data centres frequently use cutting-edge cooling technologies and renewable energy sources. Additionally, less energy is wasted on idle servers because of the effective utilisation of resources made possible by virtualisation and dynamic scaling. By relying less on energy-intensive onsite infrastructure, businesses can drastically reduce their carbon footprint by moving to the cloud. In this sense, cloud computing aids in international initiatives to promote sustainability and fight climate change.

The roles of cloud service providers

Cloud service providers (CSPs) offer a range of services over the internet, making it easier for businesses to manage their IT needs. These providers, like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud, manage the infrastructure, so you don't have to. They offer computing power, storage, and software applications that can be used on a pay-as-you-go basis. This means companies can scale up or down based on their needs without investing in physical servers or hardware.

Data storage and security: When it comes to storing data, cloud providers use massive data centres equipped with advanced technologies. Data is stored across multiple servers and often in multiple locations to ensure reliability and quick access. This setup helps in preventing data loss due to hardware failures. Security is a top priority. Providers implement various measures to protect your data, such as encryption, which means your data is turned into a code to prevent unauthorised access. They also use firewalls and other security tools to safeguard against cyber threats. Access to data is controlled by strict authentication processes, ensuring that only authorised users can view or modify it.

Cloud providers make several key promises to their customers.

- ***Scalability:*** You can easily adjust your resources based on demand without needing extra hardware.
- ***Reliability:*** They offer high uptime, meaning your services will be available nearly all the time.
- ***Security:*** They commit to protecting your data with robust security measures and compliance with industry standards.
- ***Cost-efficiency:*** You pay only for what you use, avoiding the costs of maintaining physical infrastructure.

Major cloud service providers

Amazon Web Services (AWS): AWS, launched by Amazon in 2006, is a leading cloud service provider known for its extensive range of services including computing power, storage, and databases. AWS has been a pioneer in cloud computing, offering flexible and scalable solutions that cater to various needs from startups to large enterprises. Its early entry into the market and broad service portfolio have helped it maintain a strong market position.

Microsoft Azure: Microsoft Azure, introduced in 2010, quickly became a major player in the cloud space. Azure offers a wide range of cloud services such as virtual machines,